Chapter 16, Problem 75P

Problem

When a unit step is applied to a system at $t = 0$, its response is

$$y(t) = [6 + 0.75\, e^{-3t} - e^{-2t}(3 \cos 4t + 4.5 \sin 4t)]u(t)$$

What is the transfer function of the system?

Step-by-step solution

Step 1 of 3

The transfer function of the system is,

$$H(s) = \frac{Y(s)}{U(s)}$$

The input is unit step function and its Laplace transform is,

$$U(s) = \frac{1}{s}$$

Step 2 of 3

Determine the expression for $Y(s)$.

$$Y(s) = L\left[6u(t) + 0.75e^{-3t}u(t) - 3e^{-2t}\cos 4tu(t) - e^{-2t}4.5\sin 4tu(t)\right]$$

The Laplace transform of

$0.75e^{-3t}u(t)$ Is $\frac{0.75}{s+3}$

The Laplace transform of

$e^{-2t}\cos 4tu(t)$ Is $\frac{s+2}{(s+2)^2+4^2}$

The Laplace transform of

$e^{-2t}\sin 4tu(t)$ Is $\frac{4}{(s+2)^2+4^2}$

$$Y(s) = \frac{6}{s} + \frac{0.75}{s+3} - \frac{3(s+2)}{(s+2)^2+4^2} - \frac{4.5\times 4}{(s+2)^2+4^2}$$

Step 3 of 3

Determine the expression for the transfer function of the system.

$$
\begin{aligned}
H(s) &= \frac{Y(s)}{U(s)} \\
&= \frac{\left(\frac{6}{s}+\frac{0.75}{s+3}-\frac{3(s+2)}{(s+2)^2+4^2}-\frac{4.5\times 4}{(s+2)^2+4^2}\right)}{\left(\frac{1}{s}\right)} \\
&= 6+\frac{0.75s}{(s+3)}-\frac{3s(s+2)}{(s+2)^2+4^2}-\frac{18s}{(s+2)^2+4^2}
\end{aligned}
$$

Hence the transfer function of the system is $\boxed{6+\frac{0.75s}{(s+3)}-\frac{3s(s+2)}{(s+2)^2+4^2}-\frac{18s}{(s+2)^2+4^2}}$.